

BNN Controller Change Report

Implementation summary, validation status, and remaining work

Overview

The BNN controller was repaired from a syntactically incomplete mixed-style FSM into a single-process controller with explicit state transitions. The main dataflow is now configuration, BRAM priming, activation fetch, weight fetch, compute, output commit, and completion.

Changes implemented

Area	Implemented change
FSM	Removed next_state usage and RESET state; introduced a consistent state enum with CONFIGURE_LAYER and WAIT_BRAM.
Pointers	Kept host-facing in_ptr, wt_ptr, and thr_ptr as read-only inputs; added internal active-layer pointer registers.
Syntax and declarations	Fixed the missing port comma, invalid dual state label, missing declarations, malformed layer-package logic type, and enum width/encoding conflict.
Data movement	Implemented activation buffering, per-PE weight buffering, BRAM address progression, synchronous BRAM priming, and output capture.
Output loop	Replaced the invalid NUM_PEs[i]/j loop with valid indexed procedural output capture.
Integration	Added threshold-pointer register support in the AXI-Lite wrapper at offset 0x14 and added layer_pkg.sv to the Makefile source list.
Numeric range	Increased accumulator width so the maximum 256-bit XNOR/popcount score can be represented.

Validation

Full-SoC lint completed without fatal errors. The remaining lint messages are project-wide warnings from the PicoRV32 source set and missing timescale declarations, not blocking errors in the repaired controller.

Core compilation completed successfully. The deterministic popcount-pattern regression passed. The current random and latency regressions still expose a BRAM/data-path alignment defect, and the controller-boundary tests target the old single-PE internal hierarchy and state encodings.

Remaining work

Priority	Item	Why it remains
High	Resolve BRAM data alignment	Random-vector and latency tests do not yet match the golden model.
High	Update boundary tests	Existing tests reference removed legacy signals such as dut.pe and inputs_reg.
Medium	Add programmable descriptors	layer_table exists, but no software-visible descriptor storage/load interface populates it.
Medium	Implement threshold fetch/compare	thr_ptr is configured, but no threshold BRAM port, loading logic, or decision stage exists.
Medium	Expose output vector	Only PE zero is mapped to result; output_buffer retains the other PE values internally.

Files changed: bnn_core.sv, fsm_pkg.sv, constants_pkg.sv, layer_pkg.sv, core_processing.sv, axi-lite-wraper.sv, and Makefile.